

Reirradiation of the eye with plaque brachytherapy: A single institution experience report of eight consecutive patients submitted to retreatment after local relapse of malignant disease of the eye

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ABSTRACT

PURPOSE: To evaluate the feasibility of reirradiation of the eye with plaque brachytherapy (REPBT) for local recurrence (LR) of malignant disease of the eye as an alternative to enucleation or other local salvage treatments.

METHODS AND MATERIALS: It was performed a retrospective analysis of all patients who underwent REPBT for LR. The main parameters evaluated were local control and progression-free survival, besides the toxicity profile.

RESULTS: There were eight patients who underwent REPBT, seven due to uveal melanoma and one due to retinoblastoma. The median time between the first plaque brachytherapy and the salvage plaque brachytherapy was 24 months (8–49 months). After a median followup of 30 months (8–70 months), the actuarial 2-year local control and progression-free survival was 87.5% and 60%, respectively. All patients evolved with worsening of the visual acuity and cataract. Other complications observed were maculopathy (two patients) and glaucoma (one patient).

CONCLUSIONS: REPBT should be considered as an option for salvage therapy of LR of malignant disease of the eyes as it provides a high probability of tumor control and eye preservation without compromising disease-free survival. © 2014 American Brachytherapy Society. Published by Elsevier Inc. All rights reserved.

Keywords:

Choroid melanoma; Uveal neoplasms; Retinoblastoma; Brachytherapy

Introduction

Primary malignant disease of the eye is an unusual neoplasm with an estimated incidence and mortality in 2012 in the United States of 2610 and 270, respectively (1). In adults, uveal melanoma (UM) is the most common diagnosed intraocular tumor with an annual incidence

ranging from 4.3 to 5.1 per million (2, 3). The most affected site is the choroid representing 65.15% of the cases (2). On the other hand, among children, retinoblastoma (RB) is the most frequent tumor with an estimated incidence of 360 cases in 2012 (1).

Historically, these tumors were primarily treated with enucleation. An effort has been made to evaluate eye-sparing therapies, such as plaque brachytherapy (PBT), which has a goal of improving quality of life through better cosmetic and functional results without compromising local control (LC) (4–6).

Available literature shows that conservative approach is an excellent option of treatment for selected patients with equivalent overall survival (OS) and distant metastases (DM) rates when one compares enucleation vs. PBT (7).

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